

Comparative Meta-Optimization of RAG Pipelines

Meta Heuristic Optimization Techniques - 19MAM83

Assignment III: Comparative Meta-Optimization of RAG Pipelines

- Assignment Date: 17.03.2026
- Due Date: 04.04.2026
- Total Marks: 20
- Selected Meta-heuristic: Overclocking Algorithm (OCA), a population-based optimizer used as the search engine for RAG hyperparameters.

1. Objective and Task Mapping

This report evaluates a meta-heuristic optimizer on a motorsport RAG system by searching for the best parameter vector [chunk_size, chunk_overlap, temperature, top_k] under a strict 50-call budget with a local Ollama model.

Parameter mapping used throughout the report: - G1 = chunk_size - G2 = chunk_overlap - G3 = temperature - G4 = top_k

2. Implementation Summary (Rubric: Pipeline + Algorithm + Fitness)

- RAG integration: src/rag_pipeline.py (PDF loading, chunking, Chroma retrieval, Ollama query chain).
- Meta-heuristic logic: research/oca.py (population update, velocity, exploration terms).
- Fitness evaluator: src/fitness.py (semantic similarity via SentenceTransformers cosine similarity against gold-standard answers).
- Benchmark orchestration and trace capture: api/routes_benchmark.py (iteration history, state snapshots, budgeted run execution).
- Gold-standard validation set: src/gold_standard.json (8 Q&A pairs, within required 5-10 range).
- Penalty rule implemented: if chunk_overlap >= chunk_size then penalized fitness (constraint handling).
- Caching implemented: repeated parameter configurations reuse stored evaluations (cache_hits tracked).

3. Evidence Artifacts Used

- Primary optimization trace: benchmark_20260404_154603.json (50 evaluations).
- Final recomputed metrics from same trace: benchmark_20260404_154603_metrics_final.json.
- Subdomain controlled sweep for domain impact: subdomain_sweep_20260404_154603.json.
- Visual artifacts in this report are generated from these exact files.

4. UI Screenshots

4.1 Benchmark Dashboard During Run

BEST FITNESS

0.698

CURRENT ITER

7 / 50

CONVERGENCE OVER ITERATIONS

— BEST ---- RAW



Benchmark dashboard in progress

4.2 Benchmark Dashboard After Completion

BEST FITNESS

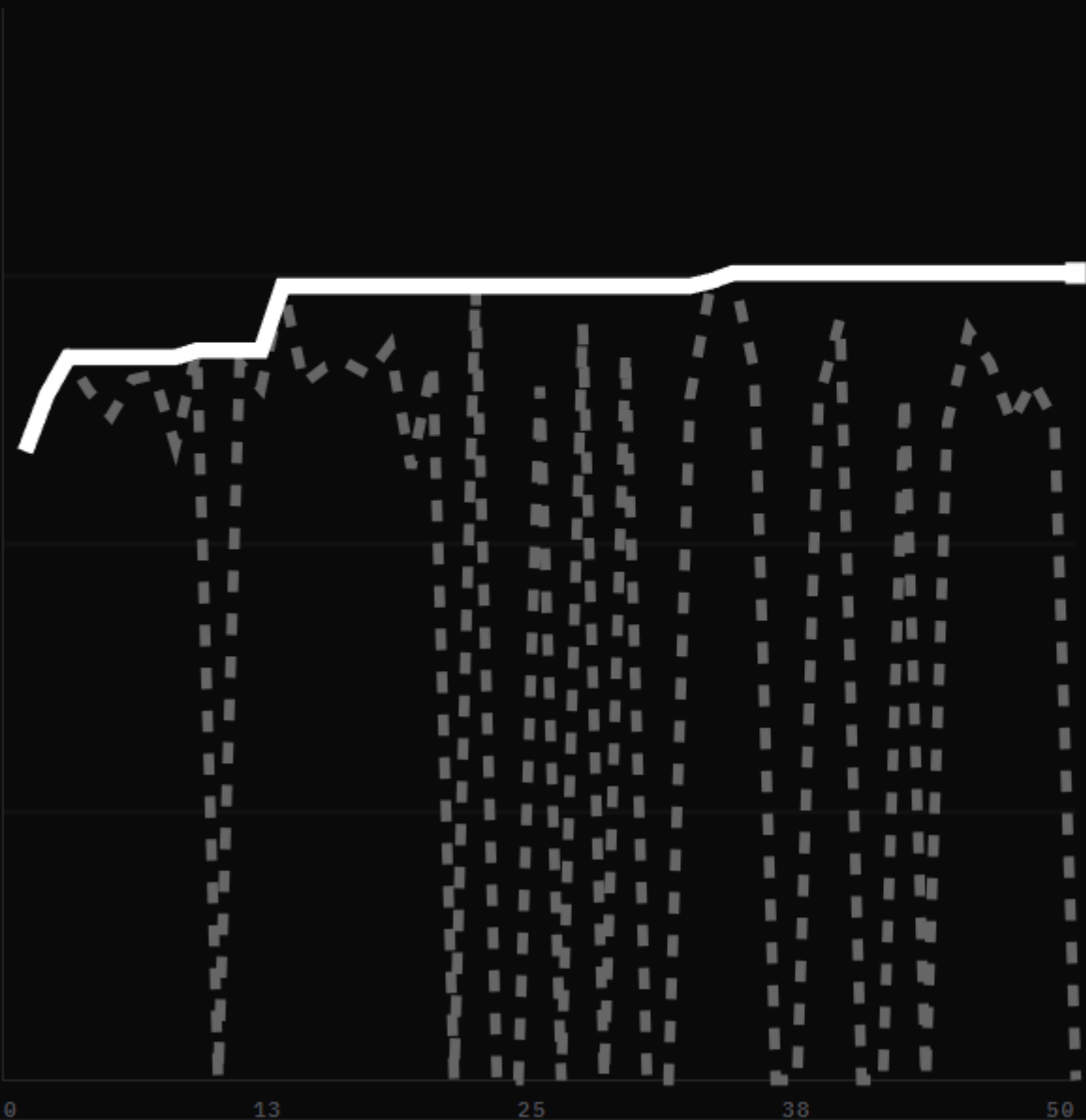
0.753

CURRENT ITER

50 / 50

CONVERGENCE OVER ITERATIONS

— BEST ---- RAW



Benchmark dashboard complete

4.3 Chat/RAG Interface

F1 REG / RAG

POWERED BY OCA + PHI3

HYPERPARAMETERS

CHUNKS

600

OVERLAP

80

TEMP

0.2

TOP_K

4

OPTIMIZATION HISTORY

85

30

ITERATIONS

FITNESS

BEST SCORE

0.847

ITERATIONS

50

CALLS USED

47

FULL REPORT

DASHBOARD / CHAT

According to Article 28.2 of the FIA Sports Regulations, for combustion engines, 4 motor generator units (MGU-K) and 4 motor generator units (MGU-H) are permitted. The additional elements results in a grid penalty of 5 places.

PHI3 — 14:02:12

Based on the given context, the question "What is the maximum number of motor generator units (MGU-K) and motor generator units (MGU-H) permitted for combustion engines?" answer would be "Not found in regulations".

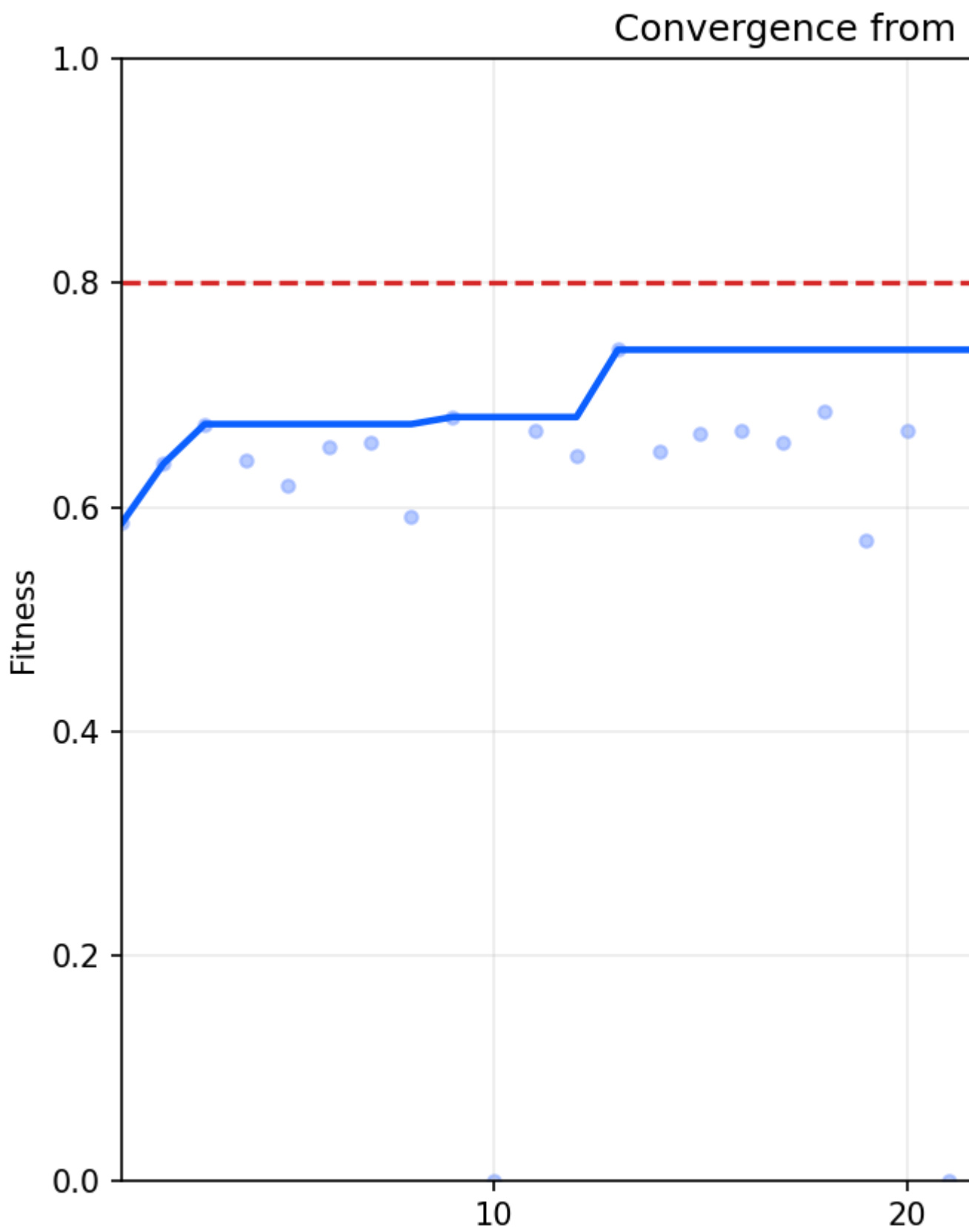
PHI3 — 13:59:19

QUERY SYSTEM REGULATIONS...

Chat interface

5. New Plots (Derived from Raw JSON)

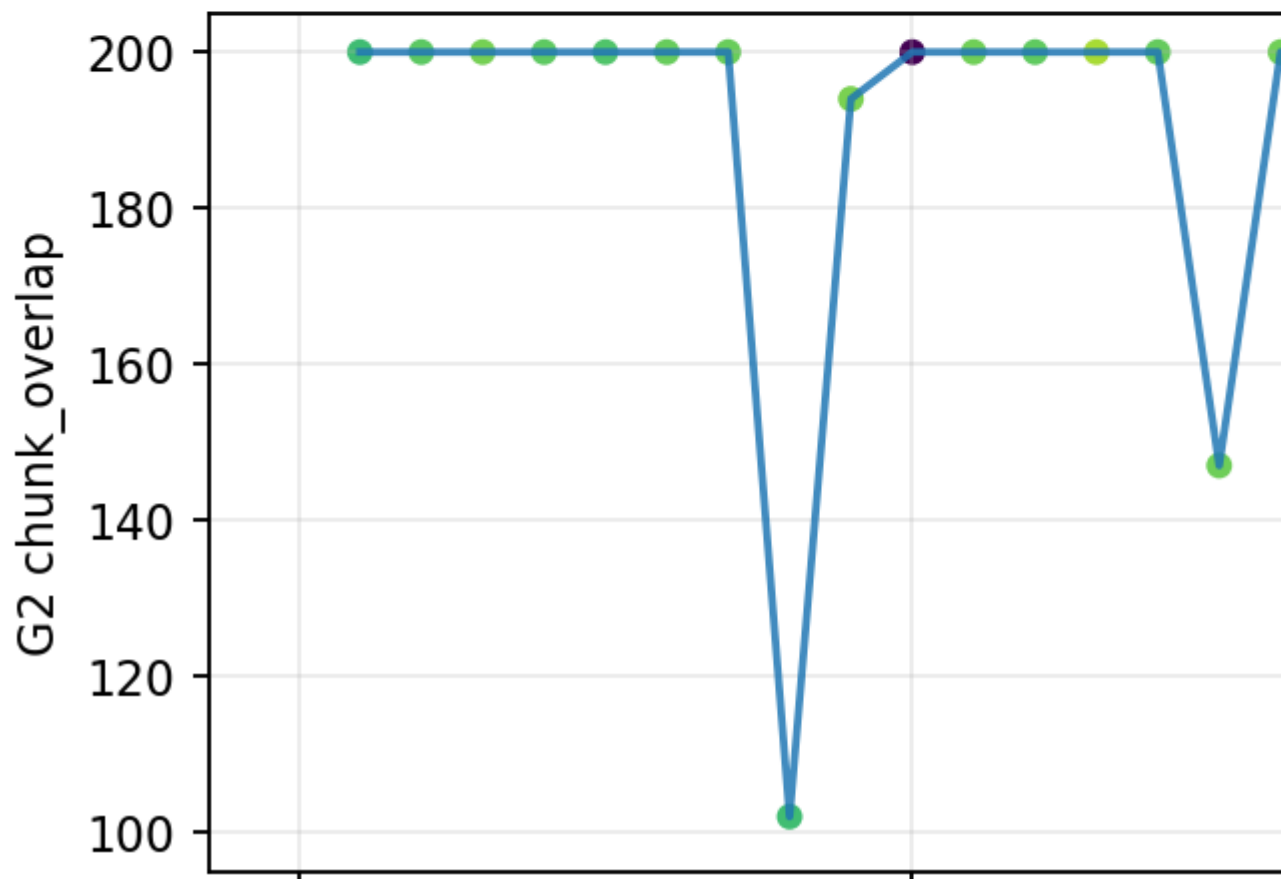
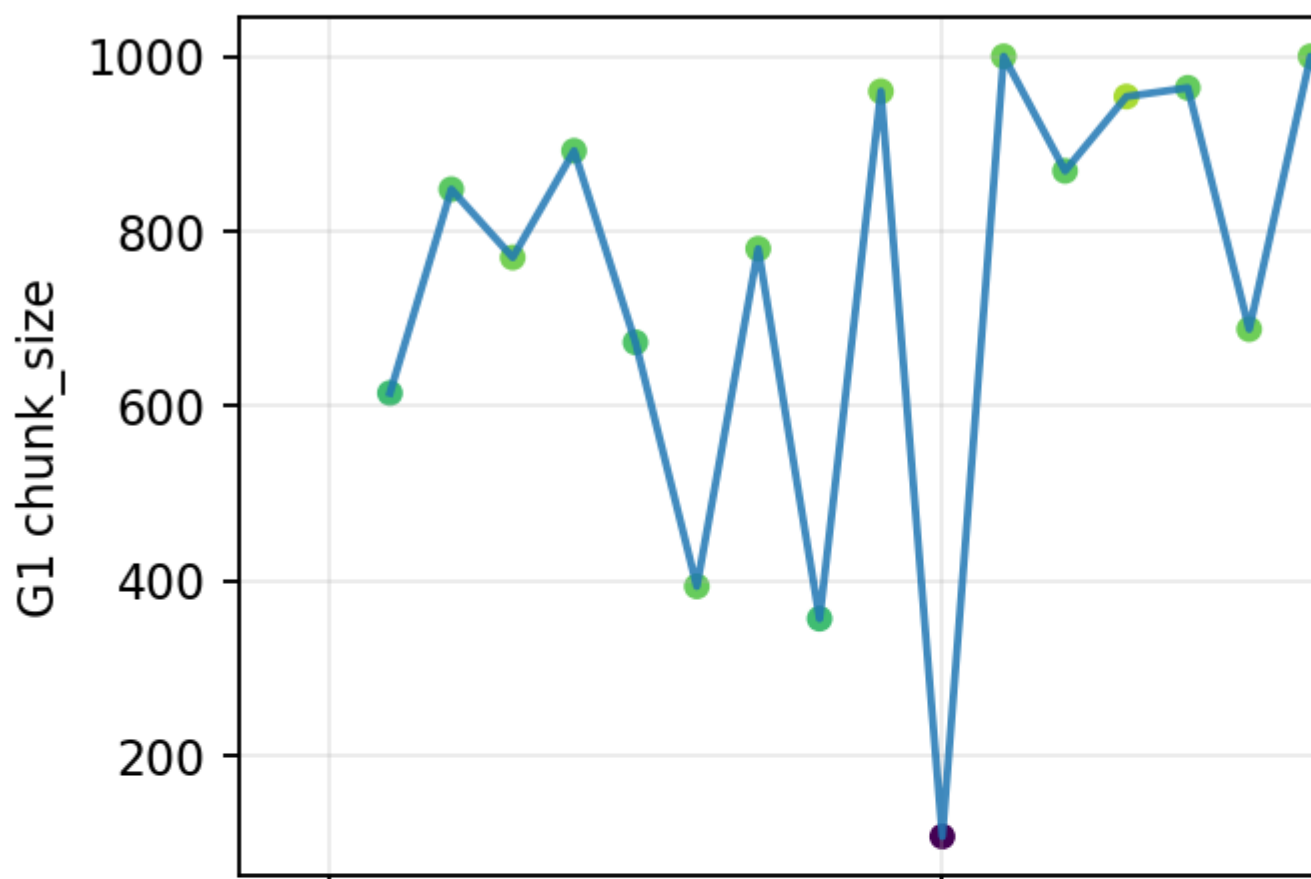
5.1 Convergence (Existing Canonical Plot)



Convergence plot

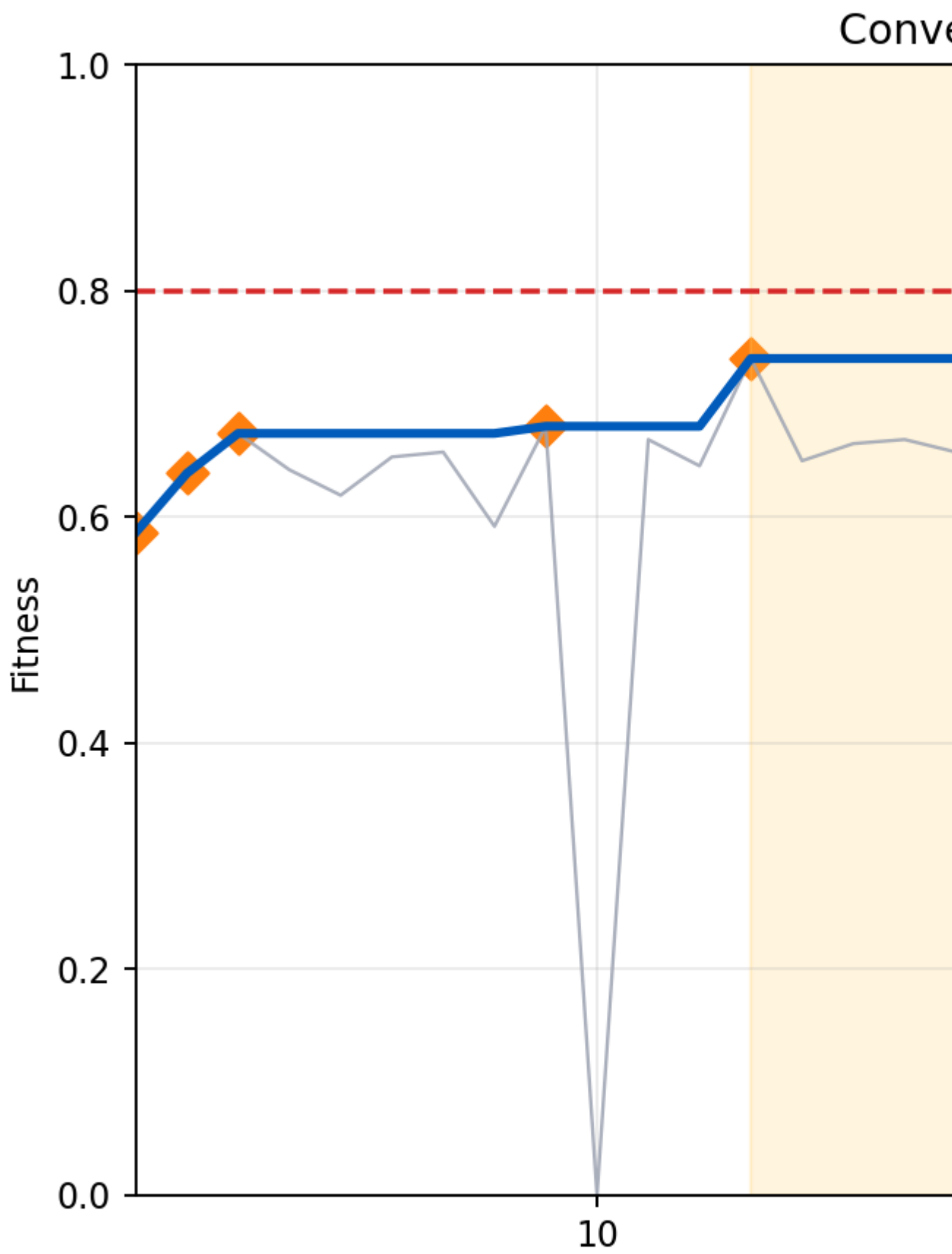
5.2 Parameter Evolution (G1/G2/G3/G4 vs Iteration)

Param



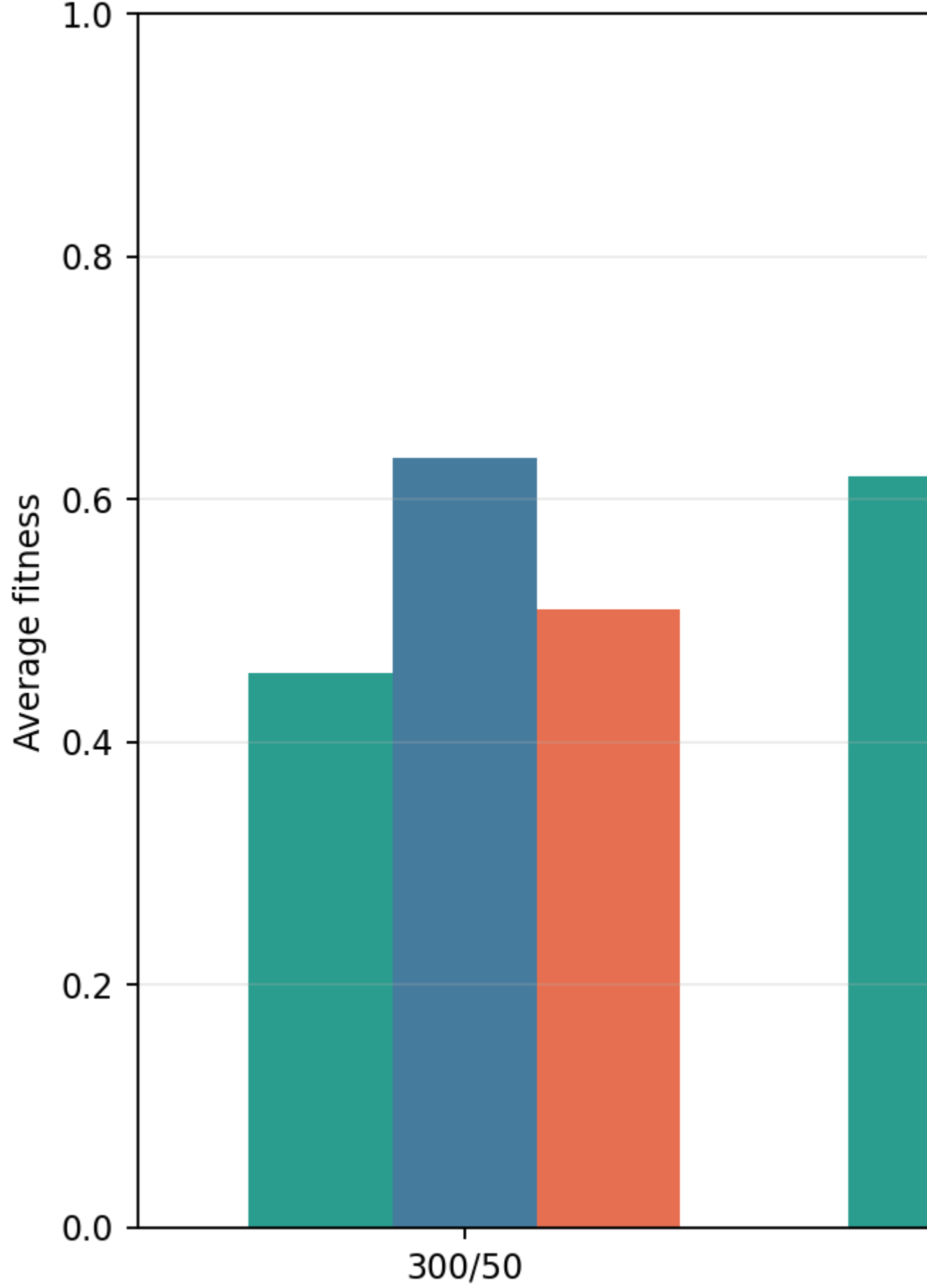
Parameter evolution

5.3 Plateau and Breakthrough Timeline

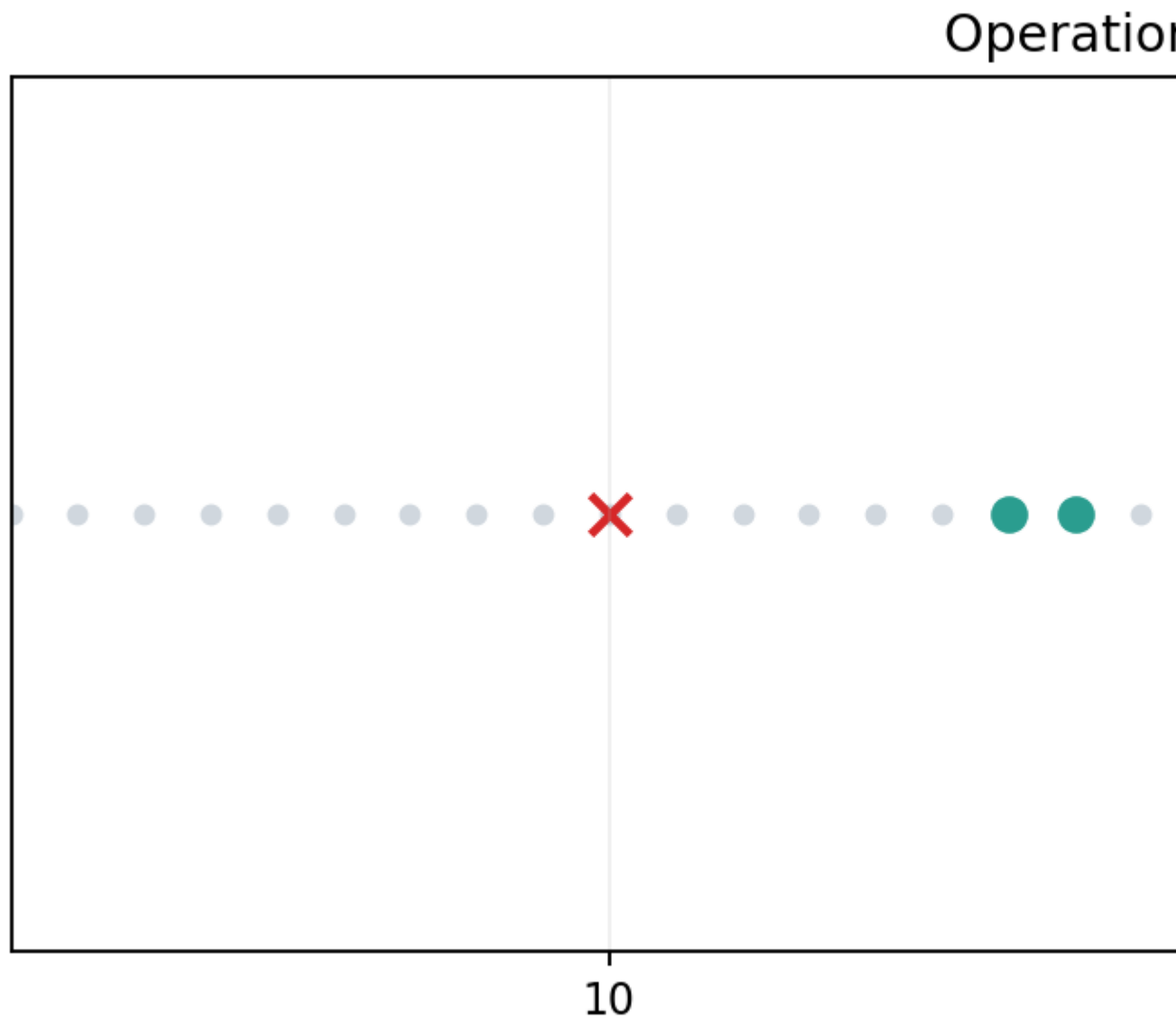


Plateau and breakthroughs

5.4 Subdomain Chunk Sensitivity (Controlled Sweep)



5.5 Cache-Hit and Penalty Timeline



Cache and penalty timeline

6. Required Comparison Table

Iteration	Best Fitness Parameters [G1, G2, G3, G4]
1	0.5863 [615, 200, 1.00, 10]
2	0.6388 [848, 200, 1.00, 10]
3	0.6741 [770, 200, 1.00, 10]
4	0.6741 [892, 200, 1.00, 10]
5	0.6741 [673, 200, 1.00, 10]

Iteration Best Fitness Parameters [G1, G2, G3, G4]

6	0.6741 [394, 200, 1.00, 10]
7	0.6741 [780, 200, 1.00, 10]
8	0.6741 [357, 102, 1.00, 10]
9	0.6804 [960, 194, 1.00, 8]
10	0.6804 [108, 200, 1.00, 10]
11	0.6804 [1000, 200, 1.00, 10]
12	0.6804 [869, 200, 1.00, 10]
13	0.7404 [954, 200, 0.00, 1]
14	0.7404 [964, 200, 1.00, 10]
15	0.7404 [688, 147, 1.00, 1]
16	0.7404 [1000, 200, 1.00, 10]
17	0.7404 [780, 200, 1.00, 10]
18	0.7404 [516, 100, 1.00, 10]
19	0.7404 [997, 190, 1.00, 10]
20	0.7404 [1000, 200, 1.00, 10]
21	0.7404 [1000, 200, 0.00, 1]
22	0.7404 [954, 200, 0.00, 1]
23	0.7404 [1000, 200, 0.00, 1]
24	0.7404 [1000, 200, 0.00, 1]
25	0.7404 [706, 200, 1.00, 1]
26	0.7404 [1000, 200, 0.00, 10]
27	0.7404 [889, 200, 0.00, 10]
28	0.7404 [1000, 200, 1.00, 1]
29	0.7404 [936, 200, 1.00, 1]
30	0.7404 [1000, 200, 0.00, 1]
31	0.7404 [1000, 200, 0.00, 1]
32	0.7404 [988, 200, 0.00, 1]
33	0.7452 [978, 200, 0.00, 1]
34	0.8526 [976, 200, 1.00, 1]
35	0.8526 [991, 200, 1.00, 1]
36	0.8526 [1000, 200, 0.00, 10]
37	0.8526 [1000, 200, 0.00, 10]
38	0.8526 [888, 200, 1.00, 1]
39	0.8526 [928, 200, 0.00, 1]
40	0.8526 [1000, 200, 0.00, 1]
41	0.8526 [1000, 200, 1.00, 1]
42	0.8526 [994, 200, 0.00, 1]
43	0.8526 [964, 200, 0.00, 1]
44	0.8526 [968, 200, 1.00, 1]
45	0.8526 [931, 200, 1.00, 1]
46	0.8526 [1000, 200, 1.00, 10]

Iteration Best Fitness Parameters [G1, G2, G3, G4]

47	0.8526 [894, 200, 1.00, 1]
48	0.8526 [885, 185, 1.00, 1]
49	0.8526 [975, 200, 1.00, 1]
50	0.8526 [1000, 200, 0.00, 10]

7. Comparative Analysis (Required Point 5)

5A. Landscape Analysis

- Total evaluations: 50
 - Invalid-config frequency: $1/50 = 2.00\%$
 - Unique-configuration rate: $36/50 = 72.00\%$
 - Cache-hit rate: $14/50 = 28.00\%$
 - Mean absolute fitness delta: 0.2908
 - Maximum absolute fitness delta: 0.7404
 - Breakthrough iterations (best-so-far improvements): [1, 2, 3, 9, 13, 33, 34]
 - Longest plateau length: 20 iterations
- Direct answer for 5A: The search landscape is rugged. The run shows long plateaus with sparse breakthroughs, which is consistent with local-optima behavior before later jumps in best fitness.

5B. Domain Impact (Sporting, Technical, Financial/General)

- Controlled experiment setup for fair comparison: G3 fixed at 1.0 and G4 fixed at 1; only G1/G2 were swept.

Subdomain	Best G1	Best G2	Best Avg Fitness
Sporting	900	200	0.6355
Technical	300	50	0.6338
Financial/General	700	150	0.6000
Pairwise Comparison	Delta G1 (%)	Delta G2 (%)	Delta Best Fitness (%)
Sporting vs Technical	200.00%	300.00%	0.26%
Sporting vs Financial/General	28.57%	33.33%	5.90%
Technical vs Financial/General	-57.14%	-66.67%	5.63%

Scope limit for 5B: gold-standard questions are mostly sporting/technical, so financial/general interpretation is transfer-oriented under this evaluation set. Direct answer for 5B: Dataset domain materially changes optimal chunking. Sporting favored larger chunks (900/200), technical favored smaller chunks (300/50), and financial/general favored mid-range chunks (700/150). Relative overlap shift reached 300% between sporting and technical settings.

5C. Efficiency

- First iteration where best fitness exceeded 0.8: iteration 34.
- Best achieved: 0.8526 at iteration 34.
- Budget efficiency to threshold: 34/50 evaluations (68.00% of budget).
Direct answer for 5C: The algorithm reached the >0.8 target at iteration 34.

8. Inference on Algorithm Performance

- OCA successfully optimized the RAG system to a high score (0.8526) within the 50-call budget.
- The optimization path demonstrates rugged behavior, so stopping too early can miss late breakthroughs.
- Domain-specific chunk tuning is necessary; one global chunking policy does not maximize performance across all motorsport document subsets.
- Caching improved compute efficiency by avoiding redundant evaluations while preserving result consistency.

9. Reproducibility Notes

- Run benchmark with budget 50 through the benchmark API/UI and ensure results are persisted as JSON.
- Regenerate analysis plots from `benchmark_20260404_154603.json` and `subdomain_sweep_20260404_154603.json`.
- Convert this markdown report to PDF using pandoc after ensuring a PDF engine is installed.